

Classic Riddler

8 April 2022

Riddle:

In the three-player Game of Mediocrity, you win by not winning too much.

Each round, every player secretly picks a number from 0 to 10. The numbers are simultaneously revealed, and the median number wins that number of points. (If two or more players pick the same number, then the winner is randomly selected from among them.)

After five rounds, the winner is whoever has the median number of points. (Again, if two or more players have the same score, then the winner is randomly selected from among them.)

With one round remaining, players A, B and C have 6, 8 and 10 points, respectively. Player A sighs and writes down “3”, but fails to do so in secret. Players B and C both see player A’s number (and both see that the other saw A’s number), and will take care to write their own numbers in secret. Assuming everyone plays to win, what numbers should B and C choose?

Solution:

By my reasoning, both B and C would choose 10. With picks of 3, 10, and 10, either B or C would get the median points for that round with 50% probability. So the final scores would be either 6, 18, and 10; or 6, 8, and 20 with 50% probability. Thus either B or C would win the game with 50% probability. I believe this is the best scenario for both B and C, because if one of them chooses 10, the other player’s best move is to also choose 10, so there is no improvement that either play can make from this scenario. So it seems to me to be a Nash equilibrium. So B and C should both choose 10.